COURSE OUTLINE

Gennoor Tech
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Automating IT and Network Operations with Claude

Eleven real operations workflows, rebuilt as supervised Claude agents

3 days (8 hours per day) | Intermediate | Instructor-led, hands-on labs | Capstone project

About this course

Who should attend

Network engineers, load-balancer and ADC administrators, NOC and IT operations teams, DevOps and platform engineers, IT service desk leads.

Prerequisites

- Working knowledge of enterprise networking and at least one load balancer (F5 BIG-IP, Citrix NetScaler or Radware)
- Comfortable on the CLI and with REST APIs and JSON
- Basic Python (reading and editing scripts); no prior AI or Claude experience required
- A ServiceNow or similar ITSM instance and a GitHub account for the labs (lab tenants are provided)

What you will be able to do

- Set up Claude Code, the Claude Agent SDK and Model Context Protocol (MCP) servers against a lab network
- Design agent workflows where Claude plans and drafts, deterministic code executes, and humans approve
- Automate certificate renewal, firmware upgrades, health checks and troubleshooting across F5, NetScaler and Radware devices
- Generate change records, tickets, runbooks, bills and flow diagrams directly from live device and mailbox data
- Ship Terraform and GitHub automation with reviews, tests, dry-runs, rollback and a full audit trail

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Tools and technologies

Claude · Anthropic Agent SDK · F5 BIG-IP · Citrix NetScaler · Radware · Cisco ACI · ServiceNow · Terraform · GitHub · GitHub Actions · Python · Ansible · Mermaid · Let's Encrypt · Docker

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DAY 1 4 modules

MODULE 1

Claude for operations engineers

- How Claude works for ops: models, context windows, structured JSON outputs and tool use
- Claude Code, the Claude Agent SDK and MCP servers as the three building blocks of every automation in this course
- The supervised-agent pattern: Claude plans and drafts, your code executes, a human approves
- Lab setup: Claude Code, Python, device simulators, ServiceNow and GitHub lab tenants

HANDS-ON LAB
Install Claude Code, connect an MCP server to a lab F5 and ask Claude to inventory virtual servers in JSON.

DELIVERABLE
Working Claude lab with your first read-only device agent.

MODULE 2

Guardrails, approvals and audit for infrastructure agents

- Least-privilege credentials, read-only by default, write actions behind explicit approval
- Dry-run, diff, pre-check and post-check as first-class steps in every workflow
- Prompt-injection and untrusted input from tickets, emails and device output
- Audit trail: what Claude saw, proposed, who approved, what ran, what changed

HANDS-ON LAB
Add an approval gate and an audit log to the inventory agent; try to trick it with a malicious ticket.

DELIVERABLE
Reusable guardrail library used by every later module.

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MODULE 3

Certificate renewal automation with Claude

- Expiry discovery across F5 BIG-IP, Citrix NetScaler, Radware and other load balancers (Anaya LB)
- CSR generation, CA request, chain validation and staged install through device APIs (iControl REST, NITRO)
- Claude-generated change tickets, maintenance windows and rollback plans
- Verification: TLS handshake checks, monitor status and automatic ticket closure

HANDS-ON LAB
Renew an expiring certificate on two different load balancers end to end, with approval and rollback.

DELIVERABLE
Multi-vendor certificate renewal agent.

MODULE 4

Device health monitoring automation with Claude

- Collect VIP status, uptime, HA state, sync state, version and last change across the estate
- Claude reads raw output and produces a normalized health report with severity and evidence
- Creating or updating ServiceNow tickets only when a state actually changes
- Scheduling, deduplication and quiet hours

HANDS-ON LAB
Build a scheduled health agent that opens a ServiceNow ticket when an HA pair falls out of sync.

DELIVERABLE
Health monitoring agent with ticket integration.

DAY 2 5 modules

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MODULE 5

Device upgrade automation with Claude

- Pre-checks: space, licences, HA state, backups, release notes read and summarized by Claude
- Disk clean-up, firmware upload, staged activation and post-checks
- Scheduler option: maintenance windows, device groups and blast-radius limits
- Automatic rollback criteria and the upgrade report

HANDS-ON LAB
Upgrade a lab HA pair with full pre- and post-checks on a schedule, then force a failed post-check.

DELIVERABLE
Upgrade orchestrator with scheduler and rollback.

MODULE 6

Cisco ACI upgrade automation with Claude

- Cisco's recommended ACI upgrade process: APIC first, then switch groups
- Fabric pre-validation with the APIC REST API and the pre-upgrade validator
- Claude-driven runbook generation, step gating and health scoring between stages
- Handling faults, pauses and resumes safely

HANDS-ON LAB
Drive a simulated ACI fabric upgrade stage by stage with Claude producing the operator runbook.

DELIVERABLE
ACI upgrade runbook agent.

MODULE 7

Troubleshooting automation from ServiceNow to CLI

- Incident lands in ServiceNow: Claude acknowledges and asks for the primary troubleshooting details
- Mapping symptoms to safe CLI command sets per platform; running them read-only
- Claude suggests next steps from the live output and keeps the ticket updated
- Templates by reason, action and next action for the knowledge base

HANDS-ON LAB
Resolve three seeded incidents using the troubleshooting agent and publish the resulting templates.

DELIVERABLE
Troubleshooting assistant with ticket workflow.

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MODULE 8

Configuration comparison across platforms

- Planning a migration: cloud-to-cloud, physical-to-cloud and cloud-to-physical
- Normalizing two configurations into one model Claude can reason about
- Semantic diff: what is equivalent, what is missing, what is risky
- Producing the migration gap report and the remediation backlog

HANDS-ON LAB
Compare a NetScaler configuration with its cloud equivalent and generate the gap report.

DELIVERABLE
Cross-platform configuration comparison tool.

MODULE 9

Traffic and flow diagram automation

- Collecting traceroute, ping and routing evidence from authorized hosts
- Claude turns raw paths into Mermaid and draw.io flow diagrams with labelled hops
- Overlaying load balancers, firewalls and NAT boundaries
- Keeping diagrams current on a schedule

HANDS-ON LAB
Generate an end-to-end flow diagram for a three-tier application from live traceroute data.

DELIVERABLE
Flow diagram generator.

DAY 3 5 modules

MODULE 10

Terraform and GitHub automation with Claude

- Terraform step by step: providers, state, modules, plan and apply, workspaces
- The common mistakes and how Claude reviews for them before plan
- Using a GitHub repository properly: branches, pull requests, reviews, protected main
- GitHub Actions pipeline that runs fmt, validate, plan and a Claude review on every PR

HANDS-ON LAB
Ship a load-balancer change as Terraform through a pull request with a Claude review and a gated apply.

DELIVERABLE
Terraform + GitHub workflow with AI review.

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MODULE 11

Credential and access management for new joiners

- Intake from HR or ticket: role, team, systems and approver
- Claude drafts access requests per channel; code raises the tickets and tracks state
- Never generating or handling secrets in the model: vault-backed credential creation
- Day-one checklist and automatic follow-ups

HANDS-ON LAB
Onboard a new joiner across four systems with tickets raised on their behalf and a tracked checklist.

DELIVERABLE
Access onboarding agent.

MODULE 12

HR hiring flow automation

- Filtering candidates from the portal of choice and downloading data
- Structured screening: English proficiency, experience, location, salary ask, preferred hours
- Call scripts and pitch generation, with human-led calls and recorded outcomes
- Fairness, consent and data-retention rules for candidate data

HANDS-ON LAB
Run a screening pipeline on a synthetic candidate set and produce the shortlist with reasons.

DELIVERABLE
Hiring screening assistant.

MODULE 13

Bill creation automation from mail

- Reading vendor and candidate mails and extracting billable items with confidence scores
- Amount checks, GST and non-GST bill variants as per company requirement
- Template rendering to PDF, numbering and ledger entry
- Exceptions queue for anything Claude is not certain about

HANDS-ON LAB
Turn a mailbox of five requests into correct GST and non-GST bills with an exceptions report.

DELIVERABLE
Bill generation agent.

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MODULE 14

Capstone: your automation, end to end

- Pick one workflow from your own environment and build it with the course guardrail library
- Demo, runbook, audit trail and a limitations statement
- Peer review and instructor feedback

HANDS-ON LAB
Capstone build and presentation.

DELIVERABLE
Production-ready automation and runbook.

Capstone

Capstone project

Choose one operations workflow from your own environment and deliver it as a supervised Claude agent with approvals, audit trail, tests, rollback and an operator runbook.

Assessment

Assessment area	Weight	Evidence
Working functionality	25%	Demonstrable end-to-end workflow in a repeatable lab
Security and safety controls	20%	Threat model, authorization boundaries, approval gates, safe failure
Testing and evaluation	20%	Automated tests, golden dataset, measured quality and known limitations
Engineering quality	15%	Clear interfaces, error handling, reproducible setup, maintainable code
Evidence and explainability	10%	Traceable findings, sources, assumptions, logs
Documentation and presentation	10%	Architecture diagram, README, runbook, demo, lessons learned

Engineering principles used throughout

- Model output is a proposal, not a fact: every finding and generated change is validated by deterministic checks and tests.
- Authorization lives outside the model: tool access is enforced by application code, scoped credentials and policy.
- Human approval for consequential actions: deploying a mitigation, changing production configuration, rotating credentials.
- Decisions, evidence, model and tool versions, approvals and outcomes are logged without recording secrets.
- A model adapter and a tool adapter layer keep every build provider-agnostic and enterprise-portable.
- Quality is measured: false positives, false negatives, reliability, latency, cost and safe failure, not just a demo.

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